EVALUATING THE FEASIBILITY OF AUTOMATED CO_2 CALCULATION USING CAD AND PDM ATTRIBUTE DATA

Lappeenranta–Lahti University of Technology LUT

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ABSTRACT

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Evaluating the feasibility of automated CO_2 calculation using CAD and PDM attribute data

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Keywords: Carbon footprint, CO_2 calculation, Scope 3 Category 1, material-based emission estimation, life cycle assessment, CAD data, PDM systems, Windchill, attribute data, data quality, data enrichment, automation, product data management, engineering data

This thesis examines the use of attribute data from engineering systems for automated CO_2 calculations. The focus is on how product data stored in CAD and product data management (PDM) systems can be used to estimate material-related emissions, and on the technical and data-related challenges associated with such automation.

The research includes a current state analysis of data structures, emission calculation practices and data quality challenges in the case company, with emphasis on the availability and reliability of mass and material data and the impact of missing mass values. Three implementation concepts were evaluated: a CAD-integrated Creo Toolkit solution, a Windchill rollup approach and a Python + Excel–based solution.

The results show that while system-integrated approaches offer the highest level of automation, their reliability depends on data completeness and consistency. Under current conditions, the Python-based approach is the most feasible short-term solution due to its flexibility and ability to handle missing data. In the long term, the Windchill rollup approach provides the most suitable solution but requires improved data quality, for example through automated data enrichment.

The findings indicate that automated emission estimation using engineering data is technically feasible but requires phased implementation, improved data quality and clear organizational ownership. Combining short-term and long-term solutions enables more data-driven and scalable carbon footprint assessments during product design.

TIIVISTELMÄ

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Konetekniikka

Mikko Toroskainen

Automatisoidun CO_2-laskennan toteutettavuuden arviointi CAD- ja PDM-attribuuttidatan avulla

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Tässä työssä tutkitaan mahdollisuuksia hyödyntää suunnittelujärjestelmien attribuuttidataa automatisoidussa hiilijalanjäljen laskennassa. Työssä tarkastellaan, kuinka CAD- ja PDM-järjestelmiin tallennettua tietoa voidaan käyttää CO_2-päästöjen arvioinnissa.

Tutkimus sisältää nykytilan analyysin, jossa tarkastellaan tietorakenteita, päästölaskennan käytäntöjä ja datan laatuun liittyviä haasteita kohdeyrityksessä. Erityistä huomiota kiinnitetään massa- ja materiaalitiedon saatavuuteen sekä puuttuvien massatietojen vaikutuksiin. Kehitystarpeiden perusteella arvioitiin kolme toteutusvaihtoehtoa: CAD-integroitu Creo Toolkit, Windchill rollup ja Python + Excel -pohjainen ratkaisu.

Järjestelmiin integroiduilla ratkaisuilla saa korkeimman automaatiotason, mutta niiden luotettavuus riippuu attribuuttidatan laadusta. Nykytilanteessa Python-pohjainen ratkaisu on soveltuvin lyhyen aikavälin vaihtoehto joustavuutensa ja puuttuvien massatietojen käsittelyn vuoksi. Pitkällä aikavälillä Windchill rollup-pohjainen laskentaratkaisua pidetään teknisesti sopivampana vaihtoehtona materiaaliperusteiseen päästöjen arviointiin. Sen käyttöönotto edellyttää kuitenkin attribuuttidatan laadun parantamista esimerkiksi automatisoidulla datan rikastuksella. Datan laadun parantuessa rollup-pohjainen laskenta mahdollistaa täysin integroidun päästölaskennan.

Tulosten perusteella attribuuttidataan perustuva automatisoitu päästöarviointi on teknisesti toteutettavissa, mutta se edellyttää vaiheittaista käyttöönottoa, datan laadun kehittämistä ja selkeää omistajuutta ympäristötiedon hallinnassa.

ABBREVATIONS

3D	Three-dimensional
BOM	Bill of materials
CAD	Computer-Aided Design
CO_2e	Carbon dioxide equivalent
ESG	Environmental, Social, and Governance
GWP	Global warming potential
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
PDM	Product Data Management
PLM	Product Lifecycle Management

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DECLARATIONS

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The author, Mikko Toroskainen, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools.

1 Introduction

The increasing focus on climate change and environmental sustainability has become a central theme in industry. Global efforts to reduce greenhouse gas emissions have made companies provide more precise and transparent reporting on the carbon footprint of their products. Policy initiatives such as the European Green Deal aim to accelerate the transition towards a low-carbon and resource-efficient economy, increasing the need for companies to better understand and manage the environmental impacts of their products. Demands from customers are also rising because expectations for environmental performance are becoming a criterion in industrial competitiveness (European Commission, 2020; European Commission, 2025). The design phase of a product has a significant impact on its environmental effects, as material choices, masses and manufacturing methods are determined in early phases of the design process. These decisions during the design process largely define the environmental burden of the product (European Commission, 2020). For this reason, assessing the CO_2 values earlier during the design process is important for internal decision-making and for external reporting.

The layout of veneer and plywood production used by Raute's customers illustrated in Figure 1, provides context for the industrial environment in which the products are designed and manufactured. The current state analysis conducted in this study, together with interviews with company representatives, indicated a need for more automated and dynamic methods for estimating and managing product emission data. Design decisions made in early engineering phases strongly influence manufacturing-related emissions, utilizing existing engineering data as a basis for preliminary emission calculations was identified as a relevant development direction. This supports the long-term development of more systematic

environmental performance assessment practices in the case company.

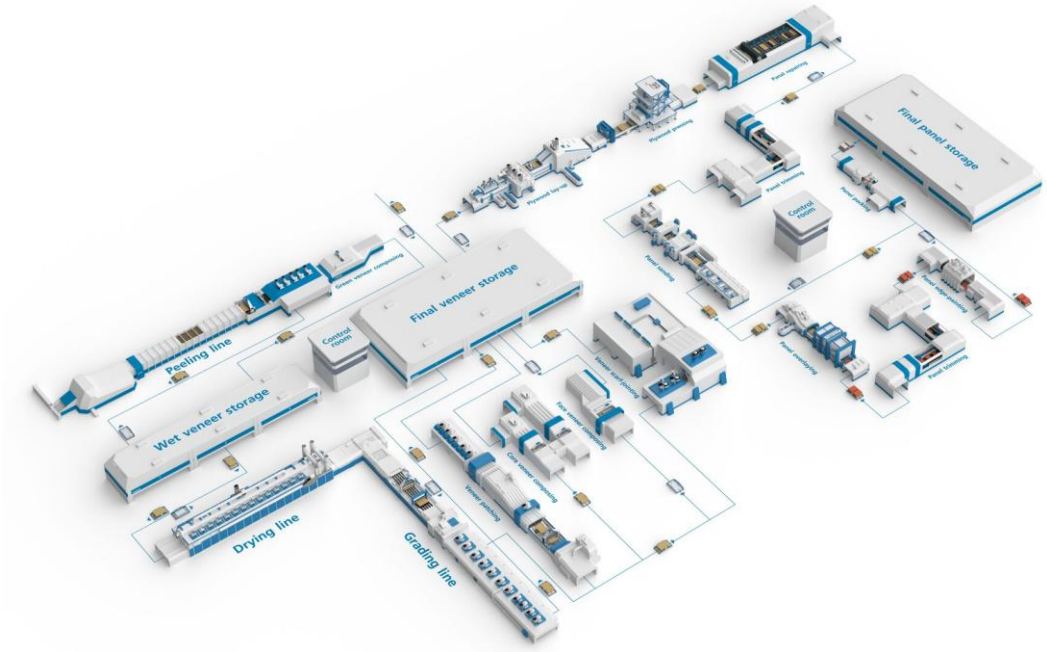


Figure 1. Raute veneer and plywood production layout (Raute design systems databank 2026).

At the case company, product development and design are based on 3D CAD models created in Creo and product data management systems such as Windchill. These systems store large amounts of engineering data, including material information, component masses and structural relationships. Such information provides a potential basis for estimating the manufacturing-related CO_2 footprint of products. Currently, product emission calculations are performed using separate assessment practices, which require manual data collection and processing. For this reason, using the existing engineering data has been identified as a potential way to support more systematic preliminary emission assessments, particularly in the context of scope 3 category 1 emissions.

The starting point of this work is the need to develop a method by which preliminary CO_2 data can be produced and calculated reliably from the data obtained from 3D CAD models. The work focuses on the phases from material acquisition to manufacturing, which excludes testing, use and end of life phases of the product. This scope limitation allows the calculation to be based on design data already available in the systems, which is continuously produced

and maintained during the process. The objective of this thesis is to develop and evaluate a concept for an automated, attribute-based emission calculation approach that utilizes CAD data to estimate material-related category 1 emissions.

1.1 Objectives of this work

The goal of this work is to find out what kind of changes to system integrations are required to produce and manage reliable preliminary CO_2 values from 3D CAD models. This information should support the design process and simplify Cat 1 reporting.

This thesis presents carbon dioxide calculator concepts that uses material data and model parameters to calculate preliminary CO_2 values related to manufacturing.

The work also examines the capability of current systems to support the generation and management of carbon footprint data. The aim is to identify potential data gaps and system limitations that hinder the effective use of the data. In addition, the study explores how preliminary emission estimation results could be integrated into product data management environments and proposes an engineering data-based tool for preliminary estimation of material-related CO_2 emission that can be linked to different systems.

Overall, this thesis aims to contribute to Raute's sustainability development by providing a structured approach for including environmental data in the early design phases.

1.2 Company background – Raute Oy

Raute is a global supplier of wood product industry technologies and services specializing in production solutions for veneer, plywood and laminated veneer lumber (LVL). Founded in 1908, Raute has grown into a globally recognized player in engineered wood production technologies. More than half of the world's LVL is produced with Raute's machines (Raute, 2025a). Raute designs and delivers complete production lines, machinery and automation solutions covering different phases of the production from log handling to panel finishing. Raute is the only company in the world that can provide complete mills (Raute, 2025b).

Raute's strategic outlook extends beyond machinery to focus on sustainability and resource efficiency. The company actively seeks to "future-proof" the wood industry by developing

solutions to increase material efficiency and minimize waste. This supports the transition toward a circular economy. This approach is guided by a comprehensive Environmental, Social, and Governance (ESG) framework. This means that Raute's technologies are designed not only to improve production performance but also lower the environmental footprint of the engineered wood value chain (Raute, 2025c).

2 Theoretical background

This chapter presents the theoretical background relevant to this thesis. This presents the key concepts, principles and technologies related to life cycle assessment, carbon footprint, emission calculation, product data, data management and system integrations. These concepts are required to define the scope and limitations of the study and to support the development of an attribute-based emission calculation approach. This chapter provides the foundation for the methods and analyses presented later in this work.

2.1 Life cycle assessment (LCA)

Life cycle assessment (LCA) is a standardized methodological framework used to quantify the potential environmental impact associated with a product, process or service throughout its life cycle. The method is based on compiling and evaluating inputs, outputs and emissions related to material and energy flows across the different stages of the product system (ISO 14040, 2006). By enabling the systematic assessment of environmental performance across the life cycle stages, LCA supports environmentally informed decision-making in product development and sustainability assessment (Hauschild et al., 2018, 61-63)

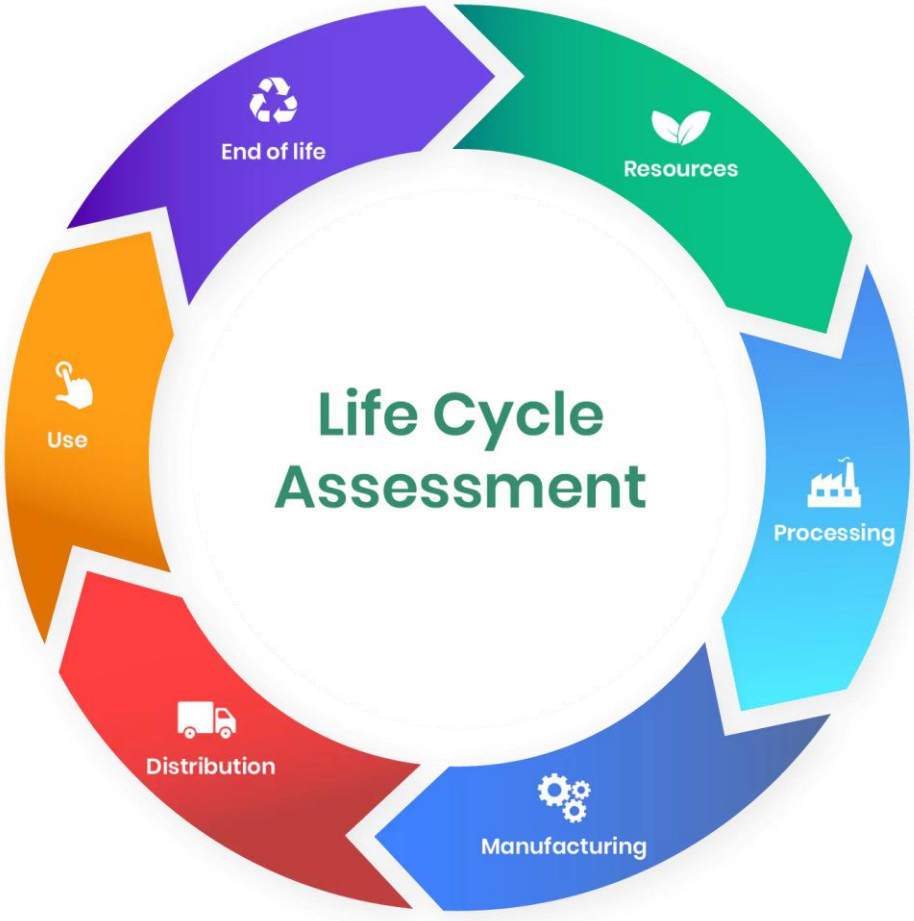


Figure 2. Phases of a life cycle assessment (OneClickLCA 2024).

From an LCA perspective, the life cycle of a product is commonly structured into stages such as raw material acquisition, manufacturing, distribution, use and end-of-life of the product (Figure 2). Depending on the objective of the assessment, system boundaries can be defined to determine which stages are included in the analysis. Commonly applied system boundary approaches include cradle-to-gate and cradle-to-grave assessments. Cradle-to-gate assessments focus on impact occurring up to the factory gate and therefore represent a more limited boundary compared to full LCA (Hauschild et al., 2018, 101-103).

LCA methodology is commonly structured to four main stages: goal and scope definition, life cycle inventory (LCI), life cycle impact assessment (LCIA) and interpretation. Goal and scope definition phase defines objectives and purpose, functional unit and the boundaries. Life cycle inventory phase focuses on collection, description and validation of data inputs, processes, emissions and other factors during products life cycle, such as material and energy

flows. Life cycle impact assessment phase, inventory data is analyzed and environmental impacts and used resources are quantified. This phase aggregates emissions into unified impact categories and may include optional normalization and weighting steps. Interpretation phase is for evaluating the results, identifying significant issues and making conclusions, limitations and recommendations (M. Sánchez-Burgos, 2025).

2.2 Carbon footprint and emission calculation principles

Carbon footprint is a commonly used indicator for quantifying the climate change impacts associated with the product, organization or activity. Corporate carbon footprint refers to the total greenhouse gas emissions generated by an organization's operations and value chain, whereas product carbon footprint focuses on emissions associated with the life cycle of a specific product. In the context of product carbon footprint assessment, it refers to the total amount of greenhouse gas emissions generated throughout the life cycle of a product, including both direct and indirect emissions related to material extraction, manufacturing, transport, use and end-of-life (ISO 14067, 2018; GHG Protocol, 2011). Product carbon footprint assessments are often derived from LCA approaches that focus specially on climate change impact rather than a broader set of environmental indicators.

To compare the climate impact of different greenhouse gases on a common basis, emissions are typically expressed using the concept of global warming potential (GWP). GWP describes the relative radiative forcing effect of a greenhouse gas compared to carbon dioxide over a set period, most commonly 100 years. Based on GWP values, the climate impacts of different gases can be expressed relative to carbon dioxide equivalents (CO_2e), which provides a standardized metric for carbon footprint calculations (IPCC, 2021).

In carbon footprint calculations, material-related greenhouse gas emissions are commonly estimated using emission factors. According to the IPCC methodology, emissions can be calculated by multiplying activity data, such as the amount of material used, with an appropriate emission factor that represents the amount of greenhouse gas released per unit of activity. Emission factors therefore express the environmental intensity of materials and processes and are typically reported as kilograms of CO_2e per unit of material. By combining material quantities with emission factors, emissions associated with material extraction,

processing and refining can be quantified in a consistent and transparent manner (GHG Protocol, 2011; Hauschild et al., 2018, 176-178).

2.3 Product data and 3D CAD models

Three-dimensional CAD models are foundation of modern engineering design and product development. CAD models contain a wide range of product-related data that describe the physical properties and relationships of components within a product. There are various attributes that include data such as geometry, mass, material definitions and assembly structures, which together represent design decisions made during the development process. These design decisions directly influence material usage, manufacturing methods and energy consumption. Such engineering data can serve as a useful starting point for emission calculations when combined with additional information, assumptions and appropriate calculation methods.

The fundamental information derived from CAD models is mass. When material density is defined, the volume of a component can be converted into mass automatically. Component mass is directly linked to the amount of material required and therefore to the environmental burden associated with material extraction, processing and transportation. Because mass is defined and its available in CAD models, it can be used as a reliable parameter for estimating material-related contributions to greenhouse gas emissions.

Another important attribute in CAD models is material information. Materials are assigned to individual parts, which links geometric models with material properties such as density and mechanical characteristics. This makes it possible to identify which materials are used in a product and in what quantities. Material data is a key input for carbon footprint calculations.

CAD environments also represent the product structure through assemblies and hierarchical relationships between components. The product structure shows how parts are combined into subassemblies and final products, including information on quantities and dependencies between components.

Together, geometry-based attributes, mass, material assignments and product structure form a comprehensive set of product data that can be used as input for carbon footprint estimation.

When product data is systematically extracted from 3D CAD models and combined with appropriate emission factors, preliminary material-related CO_2 values can be calculated during the design phase.

2.4 Product data management systems

Product data management (PDM) refers to systems and practices used to store, manage and control product-related information during engineering design and development. PDM systems provide a centralized environment for managing CAD models, drawings, documents and other product-related metadata. This helps ensure that product data is consistent, traceable and accessible to all relevant users. A key function of PDM is to support controlled handling of design information and changes, which reduce the risk of using outdated or inconsistent data (Huhtala et al., 2014).

Instead of acting only as file storage, PDM supports structured product information, such as bill of materials (BOM), relationships between components and documentation. PDM systems helps to reduce errors caused by duplicate or inconsistent data and improve the overall quality and reliability of information (Huhtala et al., 2014).

While PDM primarily focuses on managing technical data during product development, it also forms the basis for broader Product Lifecycle Management (PLM) solutions that extend data management across the entire product lifecycle (Huhtala et al., 2014). In practice, PLM enables the integration and reuse of engineering data in areas such as manufacturing planning, service management and sustainability-related activities. PDM systems play an important role in maintaining consistency between design data and product structures (Stark, 2020).

2.5 System integrations

System integration refers to connecting different information systems so that data can be exchanged consistently and automatically between them. In engineering and manufacturing environments, this typically means linking systems such as CAD, PDM, and ERP into a

shared information landscape that supports coordinated product development and lifecycle data management (Stark, 2020). In addition, the concept of a digital twin has become relevant. It enables the integration and reuse of product-related data across lifecycle stages and information systems (Tao et al., 2017).

Integration is needed because product information is typically distributed across several systems that support different phases of the product's lifecycle. In practice, product data may be created in one system, managed in another and later reused in downstream activities such as manufacturing planning and resource management. Without integration, information must be transferred manually, which can lead to inconsistencies and duplicated data. Integrated information models help to maintain consistency. They also support the coordinated use of product data across different organizational functions (Stark, 2020).

2.6 Data quality and reliability

Data quality plays a key role in all data-driven analyses and decision-making processes and is often described as data's fitness for use. In engineering and product development, the reliability of calculations, simulations and analytical assessments depends on the accuracy, completeness and consistency of the data. When input data contains errors, missing information or outdated values (timeliness), the results become misleading even when the applied methods are technically correct. For this reason, high-quality data forms the basis for reliable assessments and informed design decisions (Stark, 2020).

In product-related information systems, data quality problems often occur when information from multiple sources is combined. Heterogenous and inconsistent data structures, missing attributes and differences in data granularity across the systems reduce the usability and reliability of product information. These challenges become more pronounced in digital engineering environments where product data is reused across lifecycle stages and organizational functions (Tao et al., 2017).

Material-related data is particularly sensitive to quality issues, as many analyses rely directly on information such as material type, density and mass. In data-driven product evaluation and optimization, the availability of complete and consistently structured product data determines whether calculation-based methods can be used. When material information is

missing, incomplete or inconsistently structured, the use of product data in computational calculations or analyses becomes limited and the results become less reliable. Poorly structured material data therefore limits the effective use of engineering data in automated calculation-based assessment methods (Synnes & Welo, 2023).

3 Current state

This chapter describes the current state of product data management, system usage and emission calculation practices in case company. The purpose is to provide a clear picture of how product-related information is currently created, stored and transferred between different systems. It also examines how engineering data, such as mass and material information is handled and calculated in practice. Understanding the current state is important for identifying the main limitations and development needs that guide the proposed calculation approach. By outlining the existing data flows and working methods, this chapter establishes a baseline for evaluating the proposed solutions and improvements.

3.1 Current systems and tools

Product development and documentation in case company is supported by several systems and tools that together form the technical environment for managing product-related data. The most central systems are 3D CAD system Creo and product data management system Windchill. In addition, document management systems such as M-files and E-docs are used to store and manage product- and project-related documentation.

3.1.1 Creo

Creo is used as the primary 3D CAD tool for product design. It is used to create 3D CAD models that define the technical structure of the geometry and structure of parts and assemblies. These models serve as a source of design information during the product development process. Models include several types of attributes such as materials, mass properties calculated from geometry and material density and assembly relationships between parts and subassemblies.

The assembly structures defined in Creo represent the hierarchical product structure and define how different components and subassemblies are combined to larger assemblies. This structure determines part quantities and relationships, which are essential for generating bill of materials.

3.1.2 Windchill

Windchill functions as a product data management system for managing 3D CAD models, drawings and product data structures. It supports version control, change management and access rights. It also provides a structured environment for controlling product-related information. Product items in Windchill represent parts, assemblies, components and documents. In addition to file management, Windchill items contain attribute data such as mass and material information, classification and other metadata related to components and assemblies. In the current state, this data and its utilization for systematic environmental data management and carbon footprint calculations is limited.

3.1.3 Other systems

In addition to CAD and PDM systems, M-files is used as a document management system where technical and administrative documents related to products and projects are stored and organized. It is used for managing general project documentation as well as instructions and guidelines for different software tools and working practices, for example guidelines for E-docs publishing and system usage.

E-docs is used for storing formal technical documentation and approved product documents. It contains different kinds of engineering documents such as layout drawings, manufacturing and customer drawings, lightweight 3D models (PVZ), STEP files, schematics, cut files (DXF) and electrical design drawings. E-docs contain manufacturing-related documents and those are not connected to the parametric data or attribute data which are maintained in Creo and Windchill.

3.2 Current process for emission calculations in category 1

In the current state, emission calculations are mainly performed using Excel-based process that combines delivery information and manually collected product data. The reporting focuses on shipped product deliveries. For each delivery, the shipping department records the total mass of the delivered product to Excel file. This value is based on actual measured weight obtained from weighting during the shipment and that value is entered into the “total

mass” column in that excel. This value represents the real physical mass of the products and it is used as a reference during the material breakdown.

On the engineering side, material distribution is determined by getting the material information from Windchill for each production line. Product masses are retrieved from Windchill using separate excel worksheet, which divides mass data into different material categories. The material shares are calculated manually as percentage of the total mass, and aim is to reach full material breakdown that reaches 100%. Each row in spreadsheet represents one production line and the material distribution data is collected line by line as separate tasks. This process is time consuming, as the material distribution must be compiled individually for each delivery and there is two different excel worksheets that need to be used. The material breakdown is then used by the ESG team, which applies emission factors separately to calculate the final category 1 emissions for reporting purposes.

3.3 Data flow and challenges

The current data flow of emission calculations and reporting involves several organizational units and multiple information sources. Based on observations of existing practices and interviews, this multi-source data environment creates challenges related to data availability, consistency and interpretability. Product-related information is distributed across different systems and maintained by different organizational functions, which may lead to variation in data structures, ownership and update practices.

Challenges are observed in both logistic-related data and engineering data used for estimating material distribution and component mass. When data is generated and managed in separate processes for different purposes, achieving a consistent and comprehensive view of product-related emissions becomes more challenging.

3.3.1 Challenges in logistics-based data collection

According to interviews, collecting the total mass from logistics data can be challenging in current process. A single project ID may include a large number of different containers and

truck shipments. These shipments originate from multiple Raute sites including Nastola, Kajaani, Changzhou, Delta and Pullman, as well as subcontractors.

Data associated with a single project identifier may originate from multiple sources and may not be fully consistent in the term of how it is aggregated. As result, the available data may consist of numerous individual records, which may complicate the estimation of total mass for emission calculation purposes. As results, single project ID may involve tens to over hundred individual shipping documents that have to be manually reviewed for each total mass calculation.

3.3.2 Challenges in engineering data and material information

On the engineering side, data quality challenges are mostly related to third-party components. Parts modelled internally have accurate mass and material properties, many third-party components are missing these properties in Windchill. In these cases, the mass value is zero or blank, which limits the usability of engineering data for material or emission calculation. In addition, assemblies may contain reference components, such as foundations, which are not part of the actual product and therefore need to be excluded manually from mass calculations.

These kinds of data gaps are recognized as a realistic and unavoidable issue in complex product structures. Currently there is no standardized method for handling the missing mass data and when the data is missing or zero, the data is carried forward as such in process. There is no systematic correction or estimation method for zero values. In principle these missing values could be addressed using pre-defined assumptions, for example applying the default values based on the product categories, but no such standardized approach is currently used. With these zero or blank values, the accuracy of the material breakdown cannot be reliably assessed.

An additional challenge is related to manufacturing. Designers define the product and the structure, but suppliers manufacture the components, who can make changes based on their expert opinions. These supplier made changes doesn’t always reflect back to the engineering data, which increases the uncertainty related to the mass.

3.3.3 Mass data in engineering systems

On the engineering side, component data is primarily generated in the Creo. The mass of the part is automatically calculated based on its geometric volume and material density. When material is defined to the part, Creo uses its associated density value together with the modeled volume to calculate the mass of the component. In assemblies, those individual part mass values are calculated, and their sum becomes the total weight of the assembly. These mass values are transferred to Windchill, where they are available as part attributes and can be accessed in product structures and reports.

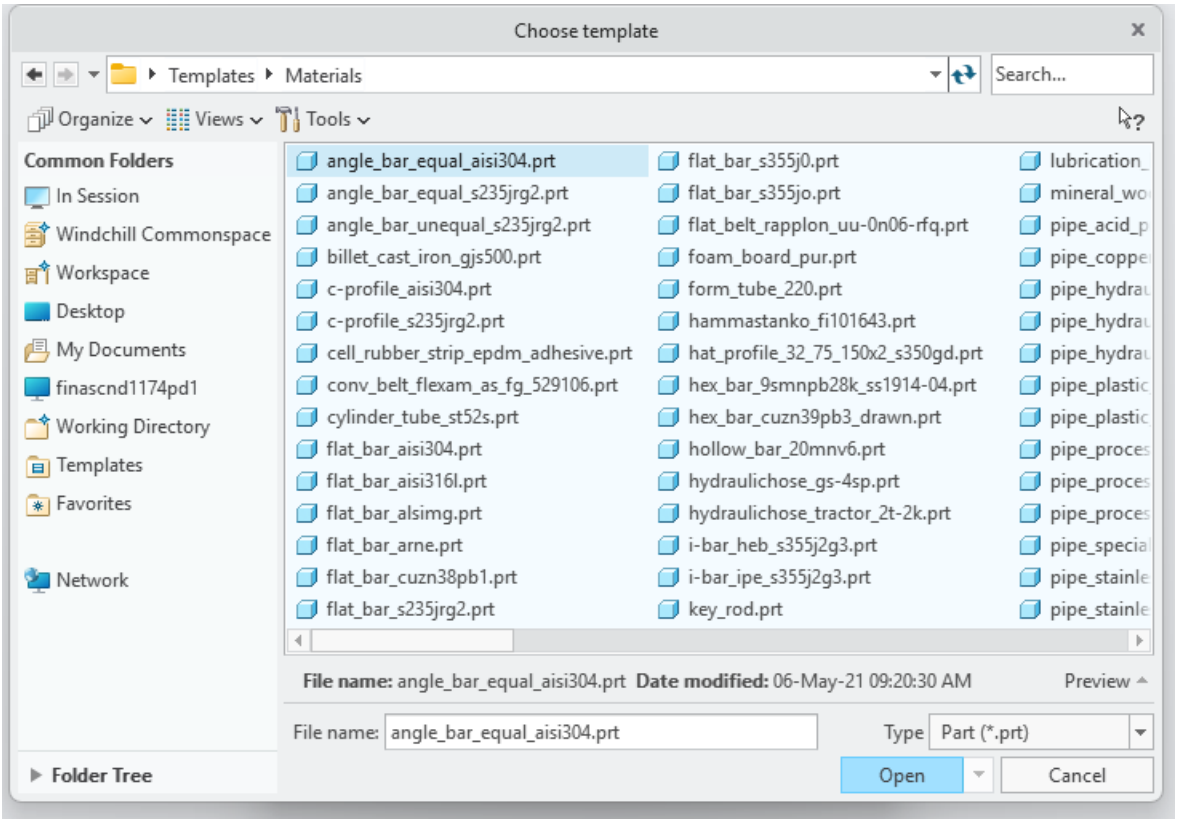


Figure 3. Start templates.

During the design process, parts are typically created using predefined start templates that have specific materials and form, such as flat bars or profiles (Figure 3). These templates have master material classes, for example steel, aluminum, cast iron or stainless steel, while template name has specific material grade, such as S355J0 or S235JRG2. Selected material

grade is stored as string parameter, and it is transferred to windchill where it is visible as part metadata. i.e. the main material is steel as numerical data in parameters, but more specific material information is in text.

From mass calculation perspective, this current approach is generally sufficient for common structural steels, as different carbon steel grades have nearly identical material densities. For this reason, using the single master material class in steel category doesn't significantly affect calculated masses. The case might be different in other material groups, such as stainless steels, where the material density differences between grades are more noticeable. For example, AISI 304 and AISI 316 have different densities, which can lead noticeable mass deviations in large or heavy components. This highlights that the master-level material classification is often sufficient for mass calculation, but its limitation becomes more relevant when the component or assembly size increases.

In addition to material classification, mass data is affected by imported third-party or purchased components. For these parts the mass information is not always available at Windchill, and the value is zero or missing entirely. This happens because for imported parts or assemblies the mass value needs to be entered manually. As result, mass data in engineering systems is partially incomplete. This limits the usability and reliability of the engineering mass data when attribute-based data is used for material breakdown analyses.

4 Development of attribute-based emission calculation

This chapter presents the development work carried out in this thesis. Based on the limitations identified in the current state analysis, there is a need to improve the usability and completeness of engineering data for emission calculation purposes. The focus is on examining how attribute data is currently visible and utilized in engineering systems, and how this data can be further developed and used to support more reliable and automated emission calculations for category 1. The current state analysis presented in chapter 3, this chapter introduces proposed improvements to attribute data usage, data completeness and calculation logic. In addition, different possible concepts for automated CO2 calculators are presented and evaluated from a practical implementation perspective.

4.1 Attribute data from engineering perspective

From the design perspective, attribute data is mainly created and maintained during the 3D modelling process. In Raute's design environment, attribute data is created and maintained during 3D modelling process in Creo, where predefined start templates define the starting point for the model and include required parameters and metadata.

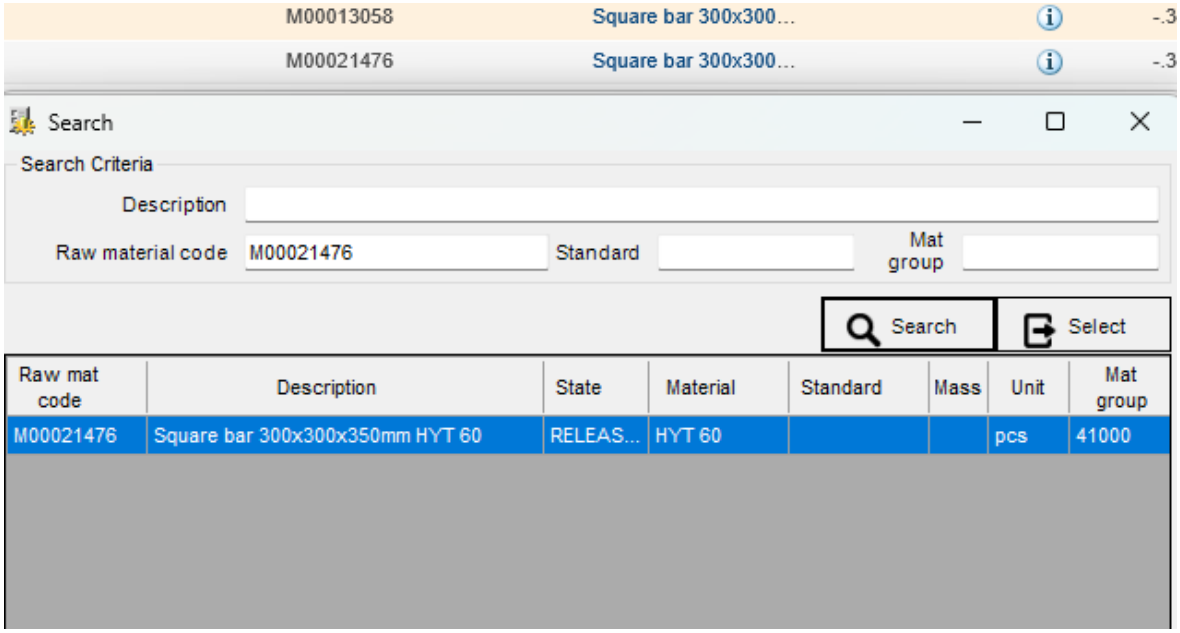


Figure 4. Inserting the material manually.

When designer starts creating a new part, the selected start template includes essential attributes such as material class and predefined parameter and relation structure. Template also provides an initial 3D geometry, such as bar or profile model, which is then modified by removing or reshaping the material to create the final part geometry. If suitable start template is not available, modeling can also start from empty. In such cases designer assign the material manually by selecting appropriate material code from Windchill, where standard material definitions are stored in a dedicated "Materials" folder. The selected material code is entered to Creo to ensure that correct density and material information are applied to the model (Figure 4). During the modelling, designers typically do not need to modify the parameter structure manually, as the required parameters are already included in the templates, and parameters like mass are automatically updated during the modelling process. The calculated mass is based on volume and density attributes.

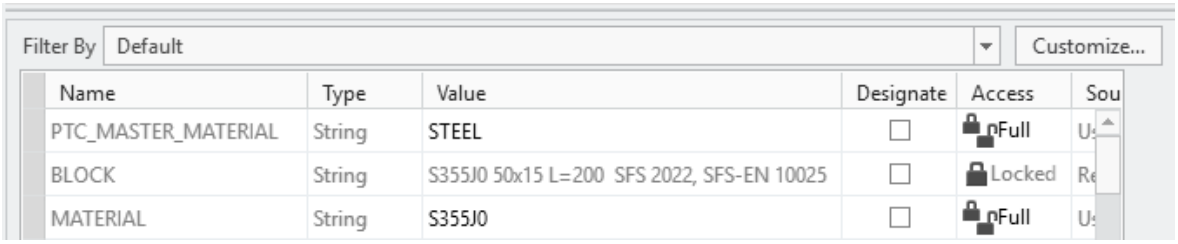
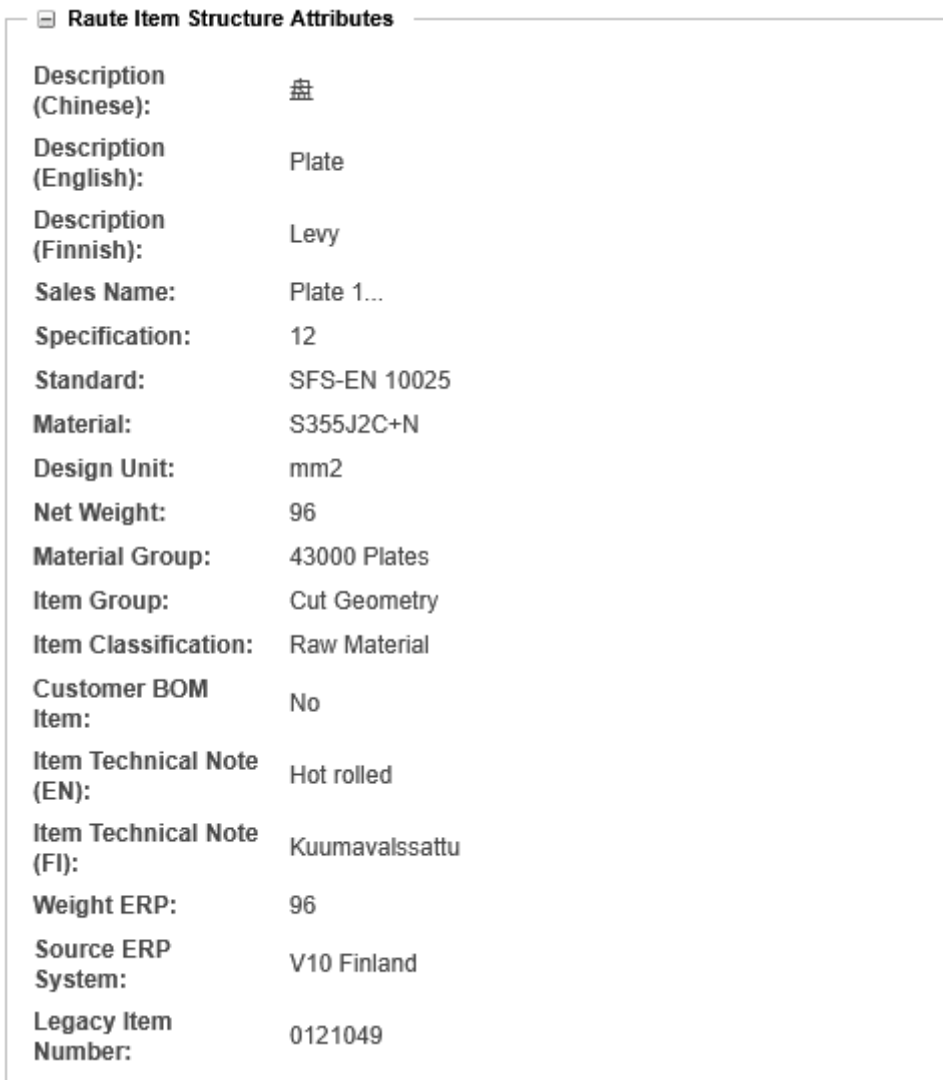


Figure 5. Material parameters.

In addition to mass-related attributes, material data is one of the most important attributes defined in Creo. Parts have master material class such as steel, stainless steel, aluminum or cast iron. More specific material grade is in the template name and it is stored as a string-based parameters named "BLOCK" and "MATERIAL" (Figure 5). This allows models to contain both numerical data needed for mass calculation and more detailed text information that can be used for identifying the actual material grade. However, string parameters are not linked to the material density used in mass calculation, which can become problematic, especially in large or heavy components where density differences between grades have a greater effect. As a result, the calculated mass is based on the master material class density rather than the specific material grade.

Creo files also stores product structure information through assemblies and subassemblies. Assemblies define how the components are connected and how many times each component appears in the final product. This product structure is important for attribute-based calculations, because it defines the quantities and relationships between components.



Raute Item Structure Attributes

Description (Chinese):	盘
Description (English):	Plate
Description (Finnish):	Levy
Sales Name:	Plate 1...
Specification:	12
Standard:	SFS-EN 10025
Material:	S355J2C+N
Design Unit:	mm2
Net Weight:	96
Material Group:	43000 Plates
Item Group:	Cut Geometry
Item Classification:	Raw Material
Customer BOM Item:	No
Item Technical Note (EN):	Hot rolled
Item Technical Note (FI):	Kuumavalssattu
Weight ERP:	96
Source ERP System:	V10 Finland
Legacy Item Number:	0121049

Figure 6. Windchill Item Structure Attributes.

Once the models are uploaded to Windchill, the attribute data from Creo is transferred into the PDM environment. In Windchill the information is visible as “Item Structure Attributes” and it can be accessed through product structures and reports (Figure 6). This enables the possible use of CAD-generated mass values and material information in downstream processes, such as exported product structure reports and calculation.

From emission calculation perspective the most important attributes available from CAD data are mass, material information and product structure. These attributes form basic dataset required for estimating material-related emissions, since material-related carbon footprint calculations are typically based on combining component masses and emission factors related to the used materials.

4.2 Identified development needs in attribute data

Based on the current state analysis presented in Chapter 3, several development needs were identified in the attribute data currently available in engineering systems. Although mass, material and product structure data exists in Creo and Windchill, the reliability and completeness of these attributes may vary depending on the component type and data origin. The limitations reduce the usability of data in automated calculations and can create uncertainty in the results.

Main issue is the missing mass data. For self-modelled parts, mass is calculated correctly in Creo, but in the case of imported third-party components and purchased parts, mass information is often missing or set to zero. As identified in Chapter 3, this creates data gaps in the product structures and limits the possibility of generating a complete material breakdown. In large assemblies, if the mass of tens to hundreds of components is unknown, the calculated mass and material shares become inaccurate and cannot be directly compared to the actual shipment weight.

Another development needs relates to the structure of material attribute data. Material information is currently stored in two forms: master material classes used for density and mass calculation, and more detailed material grade is stored as string-based parameter. String-based attributes cannot be directly used in automated calculations, because those are not numerical data. Since the material grade information does not affect density values, mass is always calculated in the same way based on master material, even though the density of the material may vary depending on the material grade. The limitations become more significant in material categories where density differences between grades are larger, and it reduce the accuracy of mass-based calculations. However, the practical impact of this limitation is currently relatively small, because majority of components are manufactured from carbon steels, where density differences between grades are minimal. In addition,

materials such as aluminum are used less frequently in Raute products, meaning that density-related deviations caused by material grade differences do not have major influence in total mass.

The use of both string-based and numeric attributes creates challenges for system integration and automation. Numeric values such as mass can be used directly in calculations, while string-based attributes require additional processing before they can be used as calculation input. In practice this would require implementing mapping logic in software, where string values (e.g. material grades) are automatically interpreted and linked to predefined numerical data such as emission factors or density values. If the name formatting of string attributes varies between models, this kind of automated interpretation becomes unreliable and requires extensive rule-based coding or manual data standardization. Because of these uncertainties, automated CO_2 calculation would also require a method to indicate the reliability of the input data and the results.

Overall, the current attribute data does not fully support automated CO_2 calculation because of missing values and inconsistent attribute formats. To enable reliable automation, attribute data should be developed towards higher completeness and consistent structure in Creo and Windchill.

4.3 Intended use of attributes in CO_2 calculation

The intended approach for automated emission calculation for category 1 based on utilizing the attribute data that is already available in engineering systems. The main idea is to use existing CAD and PDM data as calculation inputs, so that CO_2 estimates can be generated without separate manual data collection. With this approach, the calculation relies on structured attributes that describe the properties of components and assemblies.

The most important attribute inputs for this kind of calculations are mass, material information and product structure. Mass provides the numerical basis for material usage and material attributes define which emission factors should be used. Product structure defines how parts and assemblies are connected to each other and how many times each component appears in the final product. Together these attributes give a chance to apply calculation

logic where emissions are estimated at part level and then aggregated into assembly and product level results.

The intended CO_2 calculation principle is based on multiplying component mass with material-specific emission factor. Emission factors represent the greenhouse gas intensity of a material and can be expressed as kilograms of CO_2e per kilogram of material. The basic calculation format can be described as: CO_2 = Mass × emission factor.

From the system perspective, the category 1 calculation could be executed in different environments. One option is to perform the calculation directly in Creo, where parameters could be calculated and stored into models using relations or external C++ executables. Another option is to develop calculation capability further in the PDM environment, since Windchill already contains product structures, quantities and attribute data. Windchill supports rollup-type functionality, which can be used to aggregate values across the assembly structures. Another option is to calculate category 1 emissions externally using Excel and Python-based script.

When comparing the current manual approach for category 1 reporting described in Chapter 3, the implementation of automated method would reduce manual work, improve traceability and accuracy. In the current process, shipment-based weight values and material distribution are collected manually and handled separately for each delivery. An automated calculation approach would be based on engineering data that is already available in PDM system and the results could be updated dynamically when the models or product structures change. This makes the calculations faster, improves consistency between projects and support sustainability-related decision-making during the design process. In the current manual process, data collection and calculation can be time-consuming and prone to human error, especially when large product structures are processed line by line.

Overall, the use of attribute data enables systematic and repeatable CO_2 calculation, where mass and material attributes from engineering data act as primary inputs and the product structure defines aggregation logic. In practice, these kinds of automated approaches allow preliminary CO_2 estimates to be generated automatically and linked directly to engineering systems as part of product data management.

4.4 Handling missing mass and attribute data

One of the main challenges in attribute-based CO_2 calculation is handling the missing or unreliable input data. As identified in Chapters 3 and 4.2, missing mass values occur especially in third-party components, where mass values may be zero or blank in Windchill. Because mass is one the key attributes, missing values create data gaps and reduce the reliability of results. When mass values are zero, these automated calculations underestimate the total weight of the product, which leads to incorrect material breakdowns and emission estimates.

To improve the usability of engineering data, missing mass values need to be handled using predefined correction principles. One possible approach is to apply material class-based default values, where missing masses are estimated using statistical averages or median weights derived from similar components. This would allow calculation results to reach a more complete material distribution even when exact mass data is not available.

In addition to basic data correction approaches, more advanced data improvement methods could be considered in the future. One potential development direction is an external AI-supported service for improving missing mass data in Windchill. That service could identify components with missing or zero mass values in Windchill and generate estimated mass values based on existing metadata. For example, if component is under a category such as “Electrical motors”, the service could assign statistically derived average mass based on similar components. This kind of approach would significantly reduce the manual work required. Generated values could be stored as separate attributes for example “estimated mass” to maintain traceability.

4.4.1 Improving the missing data

To improve data reliability, one potential approach is to implement an automated data enrichment workflow connected to Windchill that fills the missing mass information without modifying the original CAD mass attribute, as the original values are based on geometry and material information in the CAD models and therefore represents the most accurate available

data, whereas the enriched values represents estimations. When the process starts, WTPart data is extracted from Windchill, for example through exported reports or REST/OData API queries, into a dataset containing part identifiers, mass values and classification categories. During the processing, mass units are normalized and components with valid mass values (mass > 0) are separated, while components with blank or zero values are treated as data gaps requiring estimation (Figure 7).

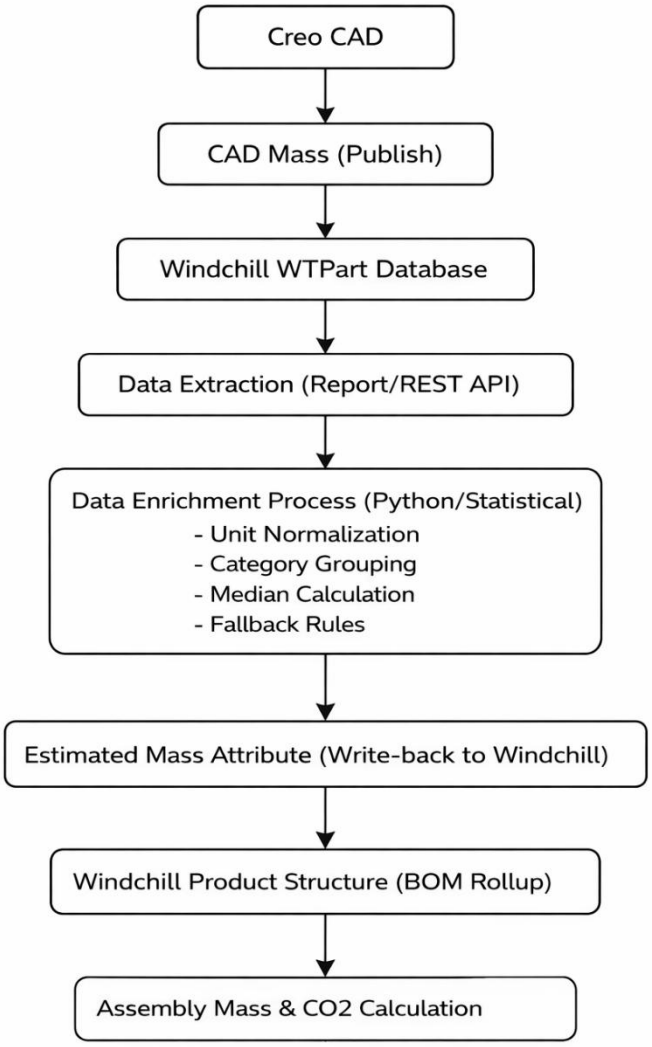


Figure 7. Mass data enrichment workflow.

Category-based reference masses can be determined by grouping components according to Windchill classifications (e.g., “Electric motors”) and calculating statistical indicators such as median mass and sample count. Median values are preferred because they reduce the influence of outliers. When components are missing mass data, an estimated value can be assigned based on its category median. If too few reference samples are available, fallback

rules can be applied for example by using higher parent classification levels, but this reduces the confidence level of the estimate. Estimated values can then be written back to Windchill as separate attributes, such as “Estimated mass”, ensuring traceability while keeping the original CAD mass values.

Once missing mass values are corrected at scale, Windchill rollup functionality can be used to generate more complete and consistent assembly-level mass totals directly from product structures. This approach provides a scalable method for improving large datasets but it requires additional system configuration and integration effort to automate data enrichment and maintain reliable attribute data over time, which can support not only emission estimation but also broader use of material-related ESG information derived from product data.

4.5 CO2 calculator concepts

This chapter presents the alternative concepts for implementing attribute-based category 1 calculations. These concepts differ in where the calculation logic is executed, how the results are stored and what kind of technical requirements and licensing are needed. The aim is to compare these concepts and highlight how each one supports automated CO_2 calculation based on engineering data.

4.5.1 Creo-based CO_2 attribute calculator using C++ Developer Toolkit (local or server-based implementation)

One potential concept for implementing an automated CO_2 calculator in Creo is to utilize C++ developer toolkit interface. With this approach, the calculation logic happens as an external C++ application that runs within Creo environment and can read and modify model parameters.

Using the C++ DevKit, the application can retrieve the active assembly, traverse its structure and identify all components and subassemblies included in the assembly. The application can access part-level parameter information such as mass properties, material related attributes and user-defined parameters. This makes it possible to automate the calculation.

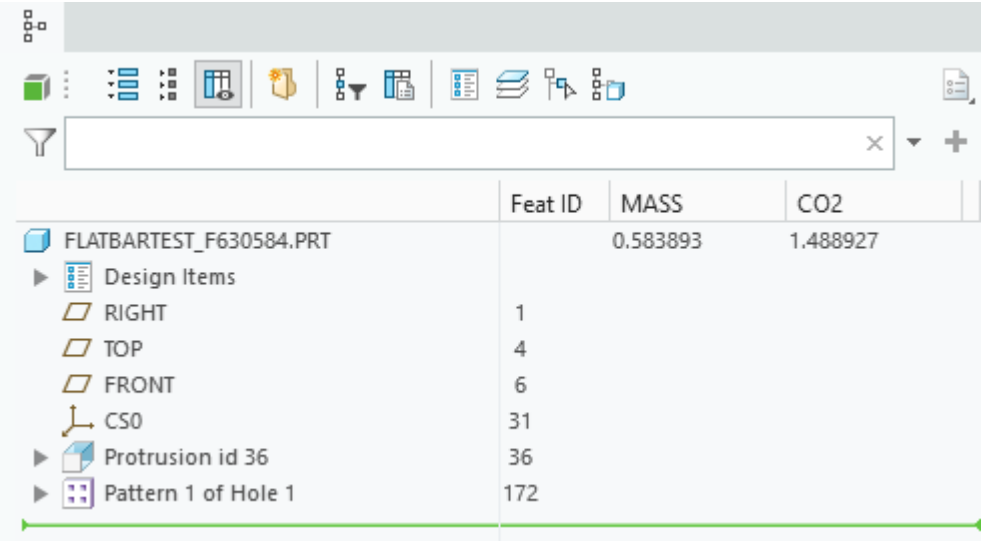


Figure 8. Model tree parameters.

The calculation workflow can be designed so that each part contains relevant parameters such as mass, emission factor and CO_2 which is calculated through relations. When processing the assembly, the application goes through entire model tree structure and determines component quantities and calculates the total mass and total CO_2 values by summing the values of all components. The results are written back to the assembly parameters such as total mass and total CO_2. These parameters can be shown directly in Creo model tree (Figure 8).

The calculation can be executed locally on designer's workstation or centralized through a dedicated server machine. In the server-based implementation, from the designer perspective, the process can be done using a simple Creo mapkey or custom button. This button generates a request file containing assembly identifier and places it on the shared network folder. Server continuously monitors the folder using watcher script. When new request is detected, server launches the toolkit-based calculator application, opens the assembly from Windchill and performs the CO_2 calculation automatically. After the calculation, the results are written into parameters, and updated models are saved back to Windchill.

The key benefit of the centralized server approach is reduced licensing requirements, since only the server machine needs the developer toolkit. In local workstation-based implementation, each designer would require the developer toolkit license, which would significantly increase the costs. With server-based approach, designers can trigger the

calculation without advanced licensing or additional software. Maintainability is also easier as updates to the logic can be done only in one location.

An important benefit of this kind of Creo-based calculator is that the results are stored directly to the CAD model data, and it is transferred to Windchill. When the parameters are in Windchill it's easy to see the CO_2 values in product structure attributes. However, the reliability of the results depends strongly on the completeness of the parameter data, especially the mass and material data, which may be missing in third-party components.

Overall, Creo toolkit-based calculator provides technically robust, flexibility and strong integration capabilities as it enables full control to the assembly parameter handling, but it also has limitations. Toolkit-based applications may be sensitive to version updates, so rebuilding and retesting might be needed after updates. This creates risk for compatibility issues over time. Another disadvantage is that large assemblies might lead to long calculation times and writing parameters straight to the CAD models requires check-out and check-in operation in Windchill, which may lead to conflicts between users.

4.5.2 Windchill rollup

One potential concept for implementing an automated CO_2 calculator is to develop the calculation logic in Windchill. This could be done by using rollup functionality in Windchill. It provides platform for calculating total masses and CO_2 values across the assemblies, since it already has attribute data from Creo. In this approach emissions could be calculated at part level using mass and material data and then summed into subassemblies and assemblies through rollup calculations. In practice, this would require developing Windchill further into calculation-supporting environment using Type and Attribute Management, where required attributes (CO_2) can be defined and maintained as part of the metadata.

Windchill provides the possibility to define calculated attributes, where stored attribute values can be derived for calculation. Windchill supports rollup calculations, which can be used to aggregate attribute values through assembly hierarchy. This makes it possible to calculate assembly-level totals by summing the part-level values. For example, if WTPart has mass parameter and CO_2 related parameter, the rollup can sum these values for assembly-

level values. This would allow the CO_2 values to be visible in Windchill without any external applications.

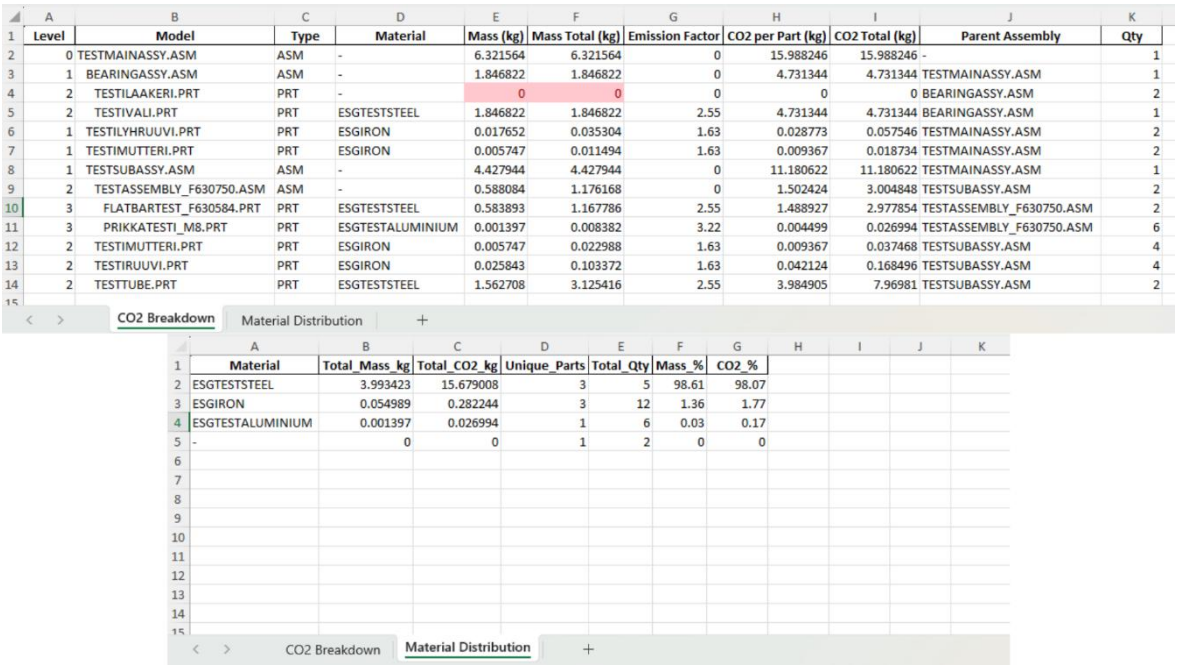
Main advantage of rollup-based calculation is centralized reporting, calculated values can be used directly in Windchill, and this can be created without manual assembly traversal. Limitation in this approach is that the results are as reliable as the input data. As identified in Chapter 3, third-party parts are often missing mass values in Windchill. Like in other approaches, if these data gaps are not addressed, total mass and CO_2 values are not 100% reliable.

Compared to the Creo Toolkit or external Python-based tool, Windchill rollup provides more centralized approach where parameter values can be used and stored directly on Windchill without external application-based approach. Windchill rollup also doesn't need to open CAD models in Creo for calculation. Windchill rollup is not as flexible and customizable as script-based approaches, and it requires that all the needed inputs already exist in Windchill.

4.5.3 Python + Excel-based calculator

One concept for implementing automated CO_2 calculation is to develop external Python-based calculation script combined with Excel-based report. From designer's perspective, the workflow could be simplified by using Creo mapkey button that triggers the calculation. When mapkey is pressed, it launches predefined script execution that runs the CO_2 calculation automatically for selected assembly and opens the generated report on the screen. This makes the process straightforward for designers without manual data handling.

The calculation principle remains the same, mass and material attributes are used as calculation inputs. CO_2 values are calculated by multiplying the component mass with predefined emission factors. Instead of performing the calculation in Creo or Windchill, the logic is implemented in a Python script. The script reads the structured model tree report exported from Creo, process the assembly hierarchy, calculate the part-level and assembly-level emissions and generate the Excel report.



Level	Model	Type	Material	Mass (kg)	Mass Total (kg)	Emission Factor	CO2 per Part (kg)	CO2 Total (kg)	Parent Assembly	Qty
0	TESTMAINASSY.ASM	ASM	-	6.321564	6.321564	0	15.988246	15.988246	-	1
1	BEARINGASSY.ASM	ASM	-	1.846822	1.846822	0	4.731344	4.731344	TESTMAINASSY.ASM	1
2	TESTILAAKERI.PRT	PRT	-	0	0	0	0	0	BEARINGASSY.ASM	2
2	TESTIVALI.PRT	PRT	ESGTESTSTEEL	1.846822	1.846822	2.55	4.731344	4.731344	BEARINGASSY.ASM	1
1	TESTILYHRUUVI.PRT	PRT	ESGIRON	0.017652	0.035304	1.63	0.028773	0.057546	TESTMAINASSY.ASM	2
1	TESTIMUTTERI.PRT	PRT	ESGIRON	0.005747	0.011494	1.63	0.009367	0.018734	TESTMAINASSY.ASM	2
1	TESTSUBASSY.ASM	ASM	-	4.427944	4.427944	0	11.180622	11.180622	TESTMAINASSY.ASM	1
2	TESTASSEMBLY_F630750.ASM	ASM	-	0.588084	1.176168	0	1.502424	3.004848	TESTSUBASSY.ASM	2
3	FLATBARTEST_F630584.PRT	PRT	ESGTESTSTEEL	0.583893	1.167786	2.55	1.488927	2.977854	TESTASSEMBLY_F630750.ASM	2
3	PRIKKATESTI_M8.PRT	PRT	ESGTESTALUMINIUM	0.001397	0.008382	3.22	0.004499	0.026994	TESTASSEMBLY_F630750.ASM	6
2	TESTIMUTTERI.PRT	PRT	ESGIRON	0.005747	0.022988	1.63	0.009367	0.037468	TESTSUBASSY.ASM	4
2	TESTIRUUVI.PRT	PRT	ESGIRON	0.025843	0.103372	1.63	0.042124	0.168496	TESTSUBASSY.ASM	4
2	TESTTUBE.PRT	PRT	ESGTESTSTEEL	1.562708	3.125416	2.55	3.984905	7.96981	TESTSUBASSY.ASM	2

CO2 Breakdown | Material Distribution

Material	Total_Mass_kg	Total_CO2_kg	Unique_Parts	Total_Qty	Mass_%	CO2_%
ESGTESTSTEEL	3.993423	15.679008	3	5	98.61	98.07
ESGIRON	0.054989	0.282244	3	12	1.36	1.77
ESGTESTALUMINIUM	0.001397	0.026994	1	6	0.03	0.17
-	0	0	1	2	0	0

CO2 Breakdown | Material Distribution

Figure 9. Python + Excel Co2 rollup output.

A key advantage of this approach is flexibility. Python enables full control for calculation logic and output structure. The output Excel could consist multiple sheets to increase usability. For example, one sheet could present assembly structure, showing quantities, total masses, materials, CO2 values and hierarchy. Second sheet could provide material distribution, showing material breakdown for total masses, total CO_2 values and percentage of the totals. Third sheet could give data reliability by summarizing zero or blank values and calculating the completeness or reliability percentage of the results (Figure 9).

Another advantage is the possibility to manage emission factors separately from engineering systems. Emission factors can be stored separate dedicated Excel file, which is read automatically during the Python script execution. This enables centralized updates of emission factors (for example using updated datasets such as Ecoinvent database) without the need for manually modifying the emission factors for each material in Creo or Windchill. In system integrated approaches, emission factors are stored in material attribute data, which may require updating material attribute values individually. By separating the emission factor management from engineering systems and using separate automatically updating Excel file, ensures that the most recent environmental data is applied in calculations. It should be noted that databases such as Ecoinvent are commercial datasets that require license.

Since the results are generated to external Excel file, the output must be stored separately to the PDM or other documentation systems. The generated report can be saved directly to M-files using the dedicated tab in Excel interface. Storing the results to Windchill requires more. The user needs to login to Windchill, locate the assembly from which the report was generated and attach the file to that product structure contents. The calculation can be automated but the integration to PDM still needs manual work.

5 Results

This chapter presents the results of the concept comparison conducted in this thesis. Based on the current state analysis and identified development needs in attribute data, several implementation approaches were examined. Comparing these approaches is necessary to identify the most suitable solution for enabling automated and scalable material-based emission calculation. This chapter compares these concepts from technical and practical perspectives, assesses their suitability within the company's system environment, and proposes a recommended approach with further development actions.

5.1 Concept comparison and suggested approach

The CO_2 calculator concepts differ significantly in terms of technical integration level, licensing requirements, flexibility and implementation complexity. Each approach is a potential option for automation, but its suitability depends on the current data quality, system architecture and organizational constraints of the case company.

5.1.1 Comparison of calculation concepts

To evaluate the identified CO_2 calculator concepts in a systematic manner, the approaches were compared using a set of technical and practical evaluation criteria. These criteria reflect the key requirements identified earlier in this thesis, such as the level of automation, integration capability, licensing implications, data reliability and maintainability. In the context of this study, the evaluation criteria are not equally weighted. Under current conditions, the most critical factors are data quality compatibility, implementation feasibility and flexibility, as the existing engineering data contains notable data gaps. Integration level and scalability become more important in the long term once the data quality has been improved. This prioritization guides the interpretation of the comparison results.

Table 1. Comparison of the CO2 concepts.

	Evaluation criteria	Creo toolkit	Windchill rollup	Python + Excel
Technical characteristics	Automation level	High (fully integrated to CAD)	Medium-high (Calculation within PDM environment)	High (Calculation automated, reporting external)
	Calculation environment	CAD environment (Creo)	PDM environment (Windchill)	External script
	System integration level	High (CAD integration, results transfers to PDM systems)	High (Native PDM level integration)	Medium (External calculation and local reports)
	Flexibility of calculation logic	High (Attribute control through C++ API)	Medium (Limited to attribute and rollup capabilities)	High (Fully customizable Python logic)
Implementation aspects	Implementation complexity	High (C++ API development and CAD integration)	Medium (Windchill configuration and attribute setup)	Medium (Python script and automation setup)
	Licensing requirements	High (Creo Developer Toolkit license required)	Medium (Windchill configuration/ customization)	Low (Python runtime + Excel)
	Maintainability	Medium (Sensitive to Creo version updates)	Medium (maintained through Windcill configuration)	High (Script updates independent from CAD systems)
Data management	Emission factor management	Manual updates (Emission factor maintained in material attributes)	Manual updates/ external database	Manual updates/ external database
	Handling missing mass data	Requires separate data correction process	Requires reliable data completeness in Windchill	Can include validation, estimation and data quality logic
	Traceability of calculation results	High (Results stored in CAD parameters and transfered to Windchill)	High (Results stored directly in Windchill attributes)	Medium (Results stored in external Excel report)
Practical Usability	Transparency of calculation outputs	High (Results visible in CAD parameters and Windchill attributes)	High (Results visible directly in Windchill structures and reports)	High (multi-sheet Excel reports)
	User accessibility	Medium (Accessible in CAD environment/ Windchill attributes)	High (Accessible in Windchill product structures/reports)	Medium-high (reports easily accessible and attachable to PDM)
	Reporting capability	Limited (basic parameter reporting within Creo)	Medium (Windchill reporting and attribute queries	High (Flexible Excel based reporting and analysis)
	User-friendliness	Medium (CAD mapkey triggering, results stored in parameter and transferred to PDM via check-in)	High (accessible through Windchill interface and reports)	Medium-high (CAD mapkey triggering, results presented in Excel report)

The comparison presented in Table 1 highlights the key trade-offs between the examined CO_2 calculation concepts in terms of technical characteristics, implementation aspects, data management and practical usability. These factors directly influence their suitability under current and future conditions. User accessibility and user-friendliness affect how easily each calculation approach can be adopted in everyday work. Although all three approaches are technically feasible, their suitability varies depending on the current system environment and the data quality limitations identified in the case company.

A Creo Toolkit-based solution provides the highest level of integration with the CAD environment and allows calculation results to be stored directly as model parameters and transferred to Windchill as part attributes. This creates a strong link between calculation results and engineering data. However, implementation requires C++ API development using the Creo Developer Toolkit and depends on the Creo environment during the execution. It also requires either a developer-level license for each user or centralized server-based architecture, which increases overall system complexity.

The Windchill rollup concept provides a centralized approach where calculations are performed within the PDM environment based on existing product structures and attribute data. This allows results to be stored and accessed directly in the product structure without the need to open the CAD models. The solution improves the accessibility of the results and supports centralized reporting. However, the reliability of the results strongly depends on the completeness and the consistency of the attribute data. As identified in Chapter 3, missing values of the third-party components currently create significant data gaps, which limits the reliability of this approach unless additional data correction or enrichment methods are implemented.

The Python + Excel-based concept provides a more flexible approach where the calculation logic is implemented outside the CAD and PDM systems while still using the engineering data as the calculation input. Compared to the system-integrated approaches, this approach enables additional validation and handling the missing mass data. This allows data quality checks, reporting of missing values and generating additional information such as material distribution and reliability information. This approach is more tolerant to the current data quality limitations.

Another important difference between these concepts relates to the required data quality level. System-integrated approaches such as Creo Toolkit and Windchill rollup rely strongly on complete and reliable attribute data. However, the current engineering data contains notable data gaps, particularly in third-party components where the mass values are often missing or zero. In contrast, the Python-based approach allows such values to be detected and handled within the calculation workflow, improving the robustness of results under current conditions.

From a practical usability perspective, approaches that require direct interaction with CAD models or specialized system configurations increases additional complexity for the user. By comparison, approaches that provide clear reporting outputs and simple execution workflows are more likely to be accepted and to support everyday engineering tasks.

Table 1 highlights that each concept has its own strengths and limitations. Python + excel-based approach offers the highest flexibility and robustness under current data quality conditions, making it the most suitable option for short-term implementation. In contrast, the Windchill rollup-based approach provides highest integration level, scalability and traceability, but requires high level of data completeness. The Creo toolkit-based approach enables highest level of CAD-level integration, but its highest implementation complexity and licensing requirements reduce its practical feasibility.

The selection of the most suitable approach depends on the balance between data quality, integration level, flexibility and implementation complexity.

5.1.2 Suggested approach for implementation

Based on concept comparison and current state analysis and the summary presented in Table 2, the most technically optimal long-term solution for automated CO_2 calculator for category 1 would be Windchill rollup-based approach, provided that the underlying attribute data quality is first improved for example with the solution presented in chapter 4.4.1. Since Windchill already manages product structures, quantities and attribute data, it offers centralized platform for aggregation-based emission calculations. In such an architecture, CO_2 values can be calculated at part level and systematically aggregated to assembly- and product-level results directly within the PDM environment. This improves accessibility of results, supports centralized reporting and enables the reuse of calculation results across different organizational functions.

Table 2. Summary of concept suitability.

Concept	Short-term suitability	Long-term suitability	Key justification
Creo Toolkit	Medium	Medium	High integration but high complexity and licensing requirements
Windchill rollup	Low	High	Requires high data quality but offers best scalability and integration
Python + Excel	High	Medium	Flexible and robust with incomplete data but limited system integration

Under current conditions in the case company, data reliability is a significant limitation for immediate implementation of this concept. As identified, missing mass values create notable data gaps in product structures. Because rollup-based aggregation depends directly on the completeness of the input attributes, these data quality issues significantly reduce the reliability of the calculated results and limit the short-term applicability of the Windchill-based solution.

For this reason, the Python + Excel-based concept is considered the most suitable solution under current conditions. This approach enables automated emission estimation by using engineering data without requiring extensive system-level customization. Calculation logic is executed outside the CAD or PDM systems, so it allows additional validation steps, handling of missing values and the generation of supporting information such as material distribution and data reliability indicators. These capabilities make this approach more tolerant to the current data limitations and improve its practical usability as an initial automation step.

Compared to the Creo Toolkit-based approach, the Python-based approach requires less complex API-level development and avoids increased licensing requirements. Although toolkit solution offers stronger interaction with CAD and enables storing the results straight

to the model parameters, its implementation is more complex, because it needs more development and maintenance effort.

Overall, the Python + Excel-based solution can be considered the most feasible short-term solution, while the Windchill rollup concept is preferred long-term system architecture once the attribute data quality challenges have been first handled.

5.2 Further action proposals

Based on the results of the concept comparison and identified limitations in data quality in engineering systems, several further development actions can be proposed to make reliable and automated CO_2 calculations in the case company.

As an initial step, it is recommended to implement the Python + Excel-based calculation approach as a practical pilot solution. This approach can be implemented with relatively low effort and without major system modifications. It enables automated emission estimations using existing engineering data and gives immediate results with material distributions, total masses, CO_2 estimates and data reliability.

After the initial implementation, systematic improvement of attribute data quality in engineering systems should be prioritized. Key development action is the creation of an automated data enrichment tool that identifies components with missing or zero values and assign estimated values based on classification-based statistical analysis. Such a solution can be implemented as a Windchill-side data processing workflow that uses historical component data to calculate the median-based masses for different material categories. By storing estimated values as separate attributes and maintaining the traceability of the data sources, the completeness and usability of the mass data are significantly improved.

In the long term, once attribute data quality has been improved through data enrichment methods, the Windchill rollup-based calculation concept can be used as a more centralized and user-friendly solution. Implementing the calculation capability directly in the PDM system enables consistent aggregation of emission-related values across product structures and allows results to be maintained in a centralized source, which can be utilized for reporting purposes.

As a further development step, the integration of a dedicated external material data management solution, such as Granta MI could be considered. Linking Windchill material attributes with externally maintained databases allows more systematic management of material parameters, such as emission factors. This reduces the need for manual updates to environmental data and improves long-term maintainability and reliability of automated Category 1 calculations.

In addition to technical implementation steps, organizational aspects also need to be considered. Clear ownership of environmental attribute data, the definition of data maintenance responsibilities and the alignment of calculation practices are important for ensuring sustainable deployment of automated CO_2 calculation methods. Periodic review of calculation results and emission factors is also necessary to maintain the reliability and usability of the generated results.

Overall, a phased development is recommended, starting with a practical pilot implementation, followed by data quality improvement through enrichment workflows and finally the implementation of a system-integrated calculation solution. This reduces implementation risks and enables a gradual transition towards automated and reliable material-based CO_2 calculation processes.

6 Conclusions

This thesis examines the possibilities of the use of attribute data from engineering systems for automated CO_2 calculations. The focus is on evaluating how existing product data stored in CAD and PDM systems can support the estimation of material-related emissions, as well as on the technical and data-related challenges associated with implementing such automation.

The results show that engineering systems already contain significant amount of relevant data for emission estimation, especially the component mass, material information and product structures. Calculation logic based on mass and emission factors provides foundation for developing automated calculation approaches. The current state analysis revealed notable limitations in data completeness, especially related to missing mass values in third-party components and the fact that some material-related information is stored as string-based attributes that cannot be directly used in automation calculations. These kinds of data quality issues currently limit the reliability of fully system-integrated CO_2 calculation solutions.

Several CO_2 calculator concepts were evaluated, including Creo Toolkit, Windchill rollup and Python + Excel-based approaches. The comparison showed that although the system-integrated approaches provide the highest level of automation and data integration, the implementation strongly depends on complete and reliable attribute data. Under current conditions in case company, the Python + Excel-based approach was identified as the most feasible short-term solution because of its flexibility, lower implementation complexity and ability to handle the missing mass data through validation and reporting mechanisms.

From a long-term perspective, the study concluded that the Windchill rollup-based approach represents the most optimal architecture for automated CO_2 estimation. However, before that can be implemented, the completeness of attribute data must first be handled. Missing or blank mass values in component attributes need to be addressed through automated data enrichment methods that estimate and rewrite the mass data based on classification-level statistical analysis. Systematic maintenance of attributes is therefore a key for achieving fully

integrated calculation workflows. In addition, integration of external material database solution could further enhance the maintainability and reliability of emission factor data.

The results of this thesis indicate that the automated CO_2 calculations based on engineering attribute data is technically achievable. At the same time, successful implementation requires both technical solutions and organizational alignment, including clear data ownership and a gradual development approach. Combining a practical short-term solution with a long-term solution allows companies to move towards more data-driven and scalable carbon footprint assessments already during the product design phase.

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